

DAY - 11 (After Mid 2)

FOURIER SERIES: (Ch#6)

Signal can be represented as a sum of its components in a variety of ways.

$$g(t) \cong c_1 x_1(t) + c_2 x_2(t) + \dots + c_N x_N(t) \\ = \sum_{n=1}^N c_n x_n(t) \quad t_1 \leq t \leq t_2$$

the Parseval's theorem says that the energy of the sum of orthogonal signal components is equal to the sum of the signal's energy.

Orthogonal Basis:-

Just as vector coordinate systems can be formed by mutually orthogonal vectors (e.g. rectangular, cylindrical, spherical), we can have signal coordinate system (basis signals) formed by a variety of mutually orthogonal signals. By definition

$$\text{Orthogonal signals } \int_{t_1}^{t_2} g(t) \cdot x(t) dt = 0$$

Two complex functions $x_1(t)$ and $x_2(t)$ are orthogonal over an interval $(t_1 \leq t \leq t_2)$

$$\int_{t_1}^{t_2} x_1(t) x_2^*(t) dt = 0 \quad \text{or} \quad \int_{t_1}^{t_2} x_1^*(t) x_2(t) dt = 0$$

we can define an orthogonal signal space as

$$\int_{t_1}^{t_2} x_m(t) x_n^*(t) dt = \begin{cases} 0 & m \neq n \\ E_n & m = n \end{cases}$$

Every signal $g(t)$ can be approximated over an

Date

interval $[t_1, t_2]$ by a set of N mutually orthogonal signals -

$$x_1(t), x_2(t), x_3(t), \dots, x_N(t)$$

there exists large number of orthogonal signals which can be used as basis signals for the generalized fourier series e.g. sinusoid functions, exponential functions, Walsh functions, Bessel functions, etc.

A periodic signal $g(t)$ can be expressed as a sum of sinusoids of frequencies $0(\text{dc}), \omega_0, 2\omega_0, \dots, n\omega_0$ whose amplitudes are $c_0, c_1, c_2, \dots, c_n$.

A famous set of orthogonal signals are the sinusoidal signals and

$$g(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$t_1 \leq t \leq t_2$$

$$\text{where } \omega_0 = \frac{2\pi}{T_0}$$

The Fourier Coefficients :-

a_0, a_n , and b_n are

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} g(t) dt$$

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} g(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} g(t) \sin n\omega_0 t dt$$

Date

the fourier series can be plotted as amplitude spectrum $[C_n \text{ vs } \omega]$ or the phase spectrum $[\phi_n \text{ vs } \omega]$ which form the frequency spectra of $g(t)$.

DAY-12

FOURIER SERIES:

$$g(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

A signal has a dual identity - time domain identity $g(t)$ and the frequency domain identity [Fourier spectra].
Two identities complement each other.

Existence of Fourier Series (Dirichlet Conditions)

1. For the series to exist, a_0 , a_n and b_n must be finite

Absolute Integrability $\leftarrow$

$$\int_{T_0} |g(t)| dt < \infty$$

weak Dirichlet conditions, Fourier series is guaranteed to exist but may not converge.

2. $g(t)$ has only finite no. of maxima and minima in one period and it may have only finite discontinuities in one period.
Combined with D $\rightarrow$ Form Strong Dirichlet conditions

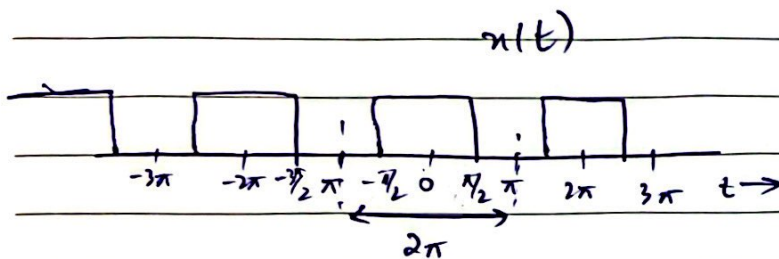
$\rightarrow$ All laboratory signals satisfy both the conditions and have convergent fourier series.

Date

Fourier representation is a way of expressing a signal in terms of ever lasting sinusoids.

Balance of magnitude and phases of components of the Fourier spectrum add up to precisely produce the pulse $g(t)$.

Question :-



Find the Fourier series for the square-pulse periodic signal
 $T_0 = 2\pi$ and $\omega_0 = \frac{2\pi}{T_0} = 1$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\pi t + b_n \sin n\pi t) \quad [\omega_0 = 1]$$

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} 1 dt \Rightarrow \frac{1}{2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos n\pi t dt = \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

$$a_n = \begin{cases} 0 & n \text{ even} \\ \frac{2}{n\pi} & n = 1, 5, 9, 13, \dots \\ -\frac{2}{n\pi} & n = 3, 7, 11, 15, \dots \end{cases}$$

$$b_n = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \sin n\pi t dt \Rightarrow 0$$

Date

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \left(\cos t - \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t - \frac{1}{7} \cos 7t + \dots \right)$$

$$A \cdot x(t) = \frac{1}{2} + \frac{2}{\pi} \left(\cos t - \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t - \frac{1}{7} \cos 7t + \dots \right)$$

DAY-13

Compact Trigonometric Fourier Series:-

When $x(t)$ is real, coefficients a_n and b_n are real for all n , and the trigonometric Fourier series can be expressed in compact form

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n)$$

where c_n and θ_n are related to a_n and b_n as

$$c_0 = a_0$$

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right)$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Periodic Signal:-

A signal $f(t)$ is said to be periodic if for some positive constant T_0

$$f(t) = f(t + T_0) \quad \text{for all } t \rightarrow \infty$$

the smallest value of T_0 which satisfies eq. (1) is called the period of $f(t)$.

A periodic signal $f(t)$

1. must be everlasting

2. $f(t)$ can be generated by periodic extension of any segment of any segment of $f(t)$ of duration T_0 (the period).

Date

3. Area under $f(t)$ over any interval of duration T_0 is same.

$f(t)$ can be expressed as

$$f(t) = C_0 + C_1 \cos(\omega_0 t + \theta_1) + C_2 \cos(\omega_0 t + \theta_2) + \dots$$

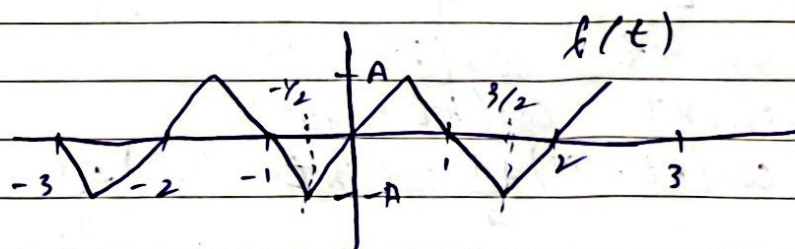
For generality, all harmonics upto infinity will be included.

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

$$f(t + T_0) = C_0 + \sum_{n=1}^{\infty} C_n \cos[n\omega_0(t + T_0) + \theta_n]$$

$$\begin{aligned} f(t + T_0) &= C_0 + \sum_{n=1}^{\infty} C_n \cos[(n\omega_0 t + n\omega_0 T_0 + \theta_n)] \\ &= C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n) \Rightarrow f(t) \end{aligned}$$

the fourier series for any even periodic function $x(t)$ consist of cosine terms only - and the series for any odd periodic function $x(t)$ consist of sine terms only.



Find the compact trigonometric fourier series for the triangular periodic signal $f(t)$ shown. sketch the amplitude and phase spectra of $f(t)$.

$$T_0 = 2$$

$$\omega_0 = \frac{2\pi}{T_0} \Rightarrow \pi$$

Date

$$f(t) = a_0 + \sum_{n=1}^{\infty} \{a_n \cos n\pi t + b_n \sin n\pi t\}$$

$$\text{where } f(t) = \begin{cases} 2At & 0 < t < 1/2 \\ 2A(1-t) & 1/2 < t < 3/2 \end{cases}$$

Average value (dc) of $f(t)$ is zero, $a_0 = 0$

$$a_n = \frac{2}{2} \int_{-1/2}^{3/2} f(t) \cos n\pi t dt$$

$$= \int_{-1/2}^{1/2} 2At \cos n\pi t dt + \int_{1/2}^{3/2} 2A(1-t) \cos n\pi t dt$$

$$a_n = 0$$

$$b_n = \int_{-1/2}^{1/2} 2At \sin n\pi t dt + \int_{1/2}^{3/2} 2A(1-t) \sin n\pi t dt$$

$$b_n = \frac{8A}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right)$$

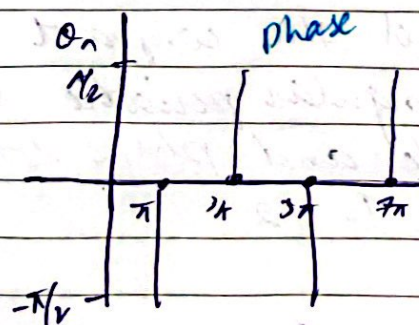
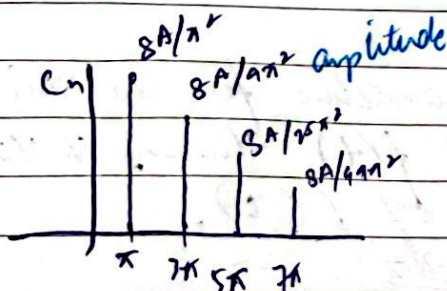
$$f(t) = \frac{8A}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \left\{ \frac{\sin(n\pi/2)}{n^2} \sin n\pi t \right\}$$

$$f(t) = \frac{8A}{\pi^2} \left[\sin \pi t - \frac{1}{9} \sin 3\pi t + \frac{1}{25} \sin 5\pi t - \frac{1}{49} \sin 7\pi t + \dots \right]$$

$$\text{Identities } \sin kt = \cos(kt - 90^\circ)$$

$$-\sin kt = \cos(kt + 90^\circ)$$

$$f(t) = \frac{8A}{\pi^2} \left[\cos(\pi t - 90^\circ) + \frac{1}{9} \cos(\sin \pi t + 90^\circ) + \frac{1}{25} \cos(5\pi t - 90^\circ) + \dots \right] \quad (\text{compact form})$$



DAY-14The Fourier Spectrum :-

The compact trigonometric Fourier series

$$x(t) = C_0 + \sum C_n \cos(n\omega_0 t + \theta_n) \rightarrow \textcircled{A}$$

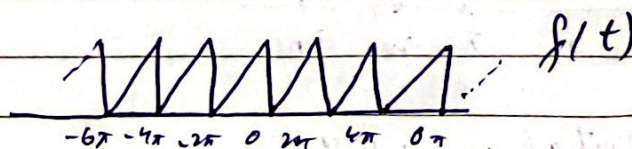
indicates that a periodic signal $x(t)$ can be expressed as a sum of sinusoids of frequencies $0(\text{dc})$, ω_0 , $2\omega_0$, ..., $n\omega_0$, whose phases are $0, \theta_1, \theta_2, \theta_3, \dots, \theta_n$ respectively.

We can readily plot C_n versus n (phase spectrum). These plots are the frequency spectra of $x(t)$.

Knowing these spectra, we can reconstruct or synthesise the signal $x(t)$ according to equation $\textcircled{A}$. The frequency spectra of a signal constitutes a frequency domain description of $x(t)$. In contrast to the time domain description, where $x(t)$ is specified as a function of time.

Question

Find the compact trigonometric Fourier series and sketch the amplitude and phase spectra of $f(t)$.

solution :-

$$T_0 = 4\pi \quad \omega_0 = \frac{2\pi}{T_0} \Rightarrow 1$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

Date

NI-YAO

$$a_0 = 0.5 \text{ (by inspection)}$$

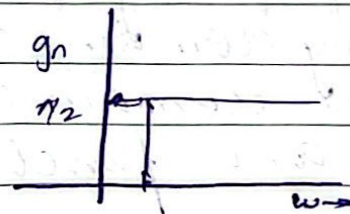
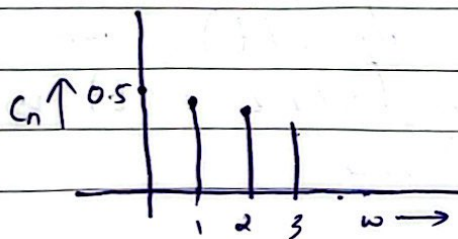
$$a_n = \frac{2}{T_0} \int_{T_0}^{t_1+T} f(t) \cos n t dt$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \frac{t}{2\pi} \cos n t dt = 0$$

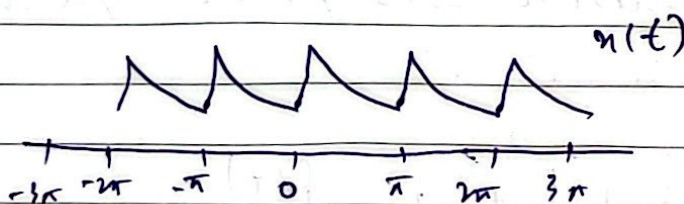
$$b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{t}{2\pi} \sin n t dt = \frac{1}{\pi n}$$

$$f(t) = 0.5 - \frac{1}{\pi} \left(\sin t + \frac{1}{2} \sin 2t + \frac{1}{3} \sin 3t + \dots \right)$$

$$f(t) = 0.5 + \frac{1}{\pi} \left[\cos\left(t + \frac{\pi}{2}\right) + \frac{1}{2} \cos\left(2t + \frac{\pi}{2}\right) + \frac{1}{3} \cos\left(3t + \frac{\pi}{2}\right) + \dots \right]$$



Question :-



the fundamental period $T_0 = \pi$

$$f_0 = \frac{1}{T_0} = \frac{1}{\pi}, \quad \omega_0 = \frac{2\pi}{T_0} = 2\pi f = 2\pi \cdot \frac{1}{\pi} \Rightarrow 2$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos 2nt + b_n \sin 2nt)$$

$$a_0 = \frac{1}{\pi} \int_{T_0} x(t) dt \Rightarrow a_0 = \frac{1}{\pi} \int_0^{\pi} e^{-t/2} dt = 0$$

Date

$$a_n = \frac{2}{\pi} \int_0^{\pi} e^{-t/2} \cos 2nt \, dt = 0.504 \left(\frac{2}{1+16n^2} \right)$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} e^{-t/2} \sin 2nt \, dt = 0.504 \left(\frac{8n}{1+16n^2} \right)$$

$$x(t) = 0.504 \left[1 + \sum_{n=1}^{\infty} \frac{2}{1+16n^2} (\cos 2nt + 4n \sin 2nt) \right]$$

$$C_0 = a_0 = 0.504$$

$$C_n = \sqrt{a_n^2 + b_n^2} = 0.504 \sqrt{\frac{4}{(1+16n^2)^2} + \frac{64n^2}{(1+16n^2)^2}}$$
$$= 0.504 \left(\frac{2}{\sqrt{1+16n^2}} \right)$$

$$\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right) = \tan^{-1} (-4n) = -\tan^{-1} 4n$$

$$x(t) = 0.504 + 0.504 \sum_{n=1}^{\infty} \frac{2}{\sqrt{1+16n^2}} \cos(2nt - \tan^{-1} 4n)$$

DAY-15

Exponential Fourier Series:

Euler's identity

$$e^{j\pi} + 1 = 0 \text{ which leads to}$$

$$e^{j\omega t} = \cos \omega t + j \sin \omega t \rightarrow \textcircled{A}$$

A periodic signal $x(t)$ can be expressed as

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega t} \rightarrow \textcircled{1}$$

Multiply $\textcircled{1}$ with $e^{-jm\omega t}$ and integrate over one period

$$\int_{T_0} x(t) e^{-jm\omega t} dt = \sum_{n=-\infty}^{\infty} D_n \int_{T_0} e^{j(n-m)\omega t} dt \rightarrow \textcircled{2}$$

Orthogonality property:-

It says

$$\int_{T_0} e^{jmt} e^{-jnw_0 t} dt = \begin{cases} 0 & m \neq n \\ T_0 & m = n \end{cases} \rightarrow \textcircled{3}$$

Use $\textcircled{3}$ in $\textcircled{2}$

$$D_n T_0 = \int_{T_0} x(t) e^{-jnw_0 t} dt \rightarrow \textcircled{4}$$

$$D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jnw_0 t} dt$$

To summarize the exponential fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jnw_0 t}$$

$$D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jnw_0 t} dt$$

Fourier Series representation of periodic signal of period T_0
 Series Form $\omega_0 = \frac{2\pi}{T_0}$

21-VAD

Trigonometric

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Compact trigonometric

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

Exponential

$$f(t) = \sum_{n=-\infty}^{\infty} D_n e^{jnw_0 t}$$

Coefficient computation

$$a_0 = \frac{1}{T_0} \int_{T_0} f(t) dt$$

Date

$$a_n = \frac{2}{T_0} \int_{T_0} f(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} f(t) \sin n\omega_0 t dt$$

$$C_0 = a_0$$

$$C_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right)$$

$$D_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-jn\omega_0 t} dt$$

Conversion Formulas

$$a_0 = C_0 = D_0$$

$$a_n - jb_n = C_n e^{j\theta_n} = 2D_n$$

$$a_n + jb_n = C_n e^{-j\theta_n} = 2D_{-n}$$

$$C_0 = D_0$$

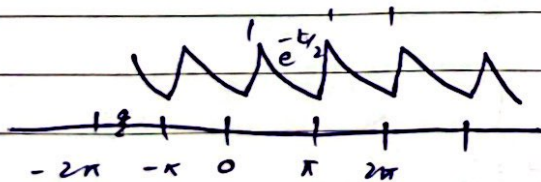
DI-KAD

$$C_n = 2|D_n|$$

$$n \geq 1$$

$$\theta_n = \angle D_n$$

QUESTION:-



Find the exponential fourier series for $x(t)$

sol.

$$T_0 = \pi$$

$$\omega_0 = \frac{2\pi}{T_0} \Rightarrow 2$$

Date

Quiz on next Monday on
trigonometric

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{j2\pi n t}$$

where

$$\begin{aligned} D_n &= \frac{1}{T_0} \int_{T_0} x(t) e^{-j2\pi n t} dt = \frac{1}{\pi} \int_0^{\pi} e^{t/2} e^{-j2\pi n t} dt \\ &= \frac{1}{\pi} \int_0^{\pi} e^{-(1/2 + j2\pi n)t} dt = \frac{-1}{\pi(1/2 + j2\pi n)} e^{-(1/2 + j2\pi n)t} \Big|_0^{\pi} \end{aligned}$$

$$D_n = \frac{0.504}{1 + j4n}$$

$$\begin{aligned} x(t) &= 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2\pi n t} \\ &= 0.504 \left[1 + \frac{1}{1 + j4} e^{j2t} + \frac{1}{1 + j8} e^{j4t} + \dots + \frac{1}{1 - j4} e^{-j2t} + \frac{1}{1 - j8} e^{-j4t} + \dots \right] \end{aligned}$$

DAY-16

Exponential Fourier Series:

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{j2\pi n t} \quad \text{--- Generic exponential Fourier Series}$$

$$\begin{aligned} x(t) &= 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2\pi n t} \\ &= 0.504 \left[1 + \frac{1}{1 + j4} e^{j2t} + \frac{1}{1 + j8} e^{j4t} + \dots + \frac{1}{1 - j4} e^{-j2t} + \frac{1}{1 - j8} e^{-j4t} + \dots \right] \end{aligned}$$

Observe that the coefficients D_n are complex. Moreover, D_n and D_{-n} are conjugate.

Date

For the exponential fourier series, we plot the magnitude $|D_n|$ and the angle of D_n $\angle D_n$.

this requires that the coefficients D_n be expressed in polar form as

$$|D_n| e^{j\angle D_n}$$

For the above series

$$D_0 = 0.504$$

$$D_1 = \frac{0.504}{1+j4} = 0.122 e^{-j75.96^\circ} \Rightarrow |D_1| = 0.122$$

$$\angle D_1 = -75.96^\circ$$

$$D_{-1} = \frac{0.504}{1-j4} = 0.122 e^{j75.96^\circ} \Rightarrow |D_{-1}| = 0.122$$

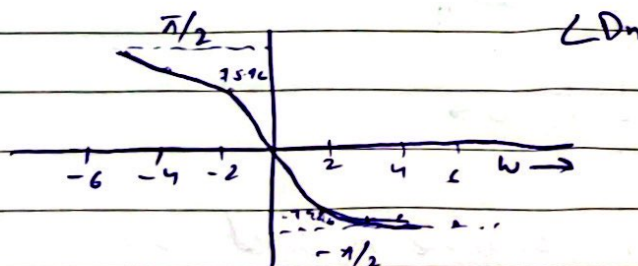
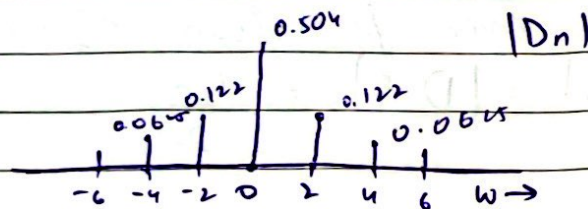
$$\angle D_{-1} = 75.96^\circ$$

$$D_2 = \frac{0.504}{1+j8} = 0.0625 e^{-j82.87^\circ} \Rightarrow |D_2| = 0.0625$$

$$\angle D_2 = -82.87^\circ$$

$$D_{-2} = \frac{0.504}{1-j8} = 0.0625 e^{j82.87^\circ} \Rightarrow |D_{-2}| = 0.0625$$

$$\angle D_{-2} = 82.87^\circ$$

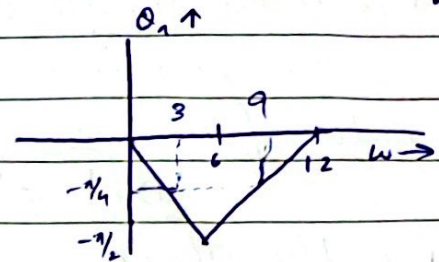
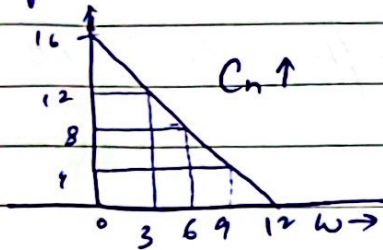


QUESTION ::

The trigonometric fourier series is shown - sketch the

Date

exponential fourier series and verify result analytically



The trigonometric fourier series has four spectral components at w freq. 0, 3, 6 and 9.

the DC component is 16.

phase component of w freq. 3 and 9 is $(-\frac{\pi}{4})$

phase component of w freq. 6 is $-\frac{\pi}{2}$

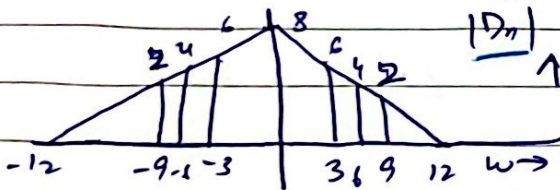
$$x(t) = 16 + 12\cos(3t - \pi/4) + 8\cos(6t - \pi/2) + 4\cos(9t - \pi/4)$$

For the exponential fourier series

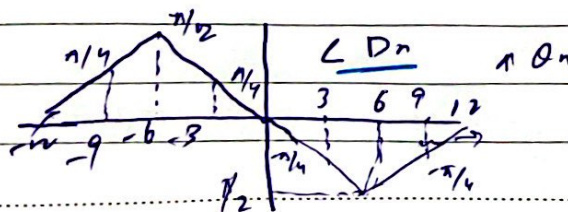
$$D_0 = C_0 = 16$$

$|D_n|$ is an even function of w

$$|D_n| = |D_{-n}| = \frac{C_n}{2}$$



$\angle D_n = \theta_n$ for positive n and $-\theta_n$ for negative n



Date

$$\hat{n}(t) = 16 + \left[\frac{6 e^{-j\pi/4} e^{j3t}}{4 e^{j\pi/2} e^{-j6t}} + \frac{6 e^{j\pi/4} e^{-j3t}}{4 e^{-j\pi/2} e^{j6t}} \right] + \left[\frac{2 e^{-j\pi/4} e^{j9t}}{2 e^{j\pi/4} e^{-j4t}} + \frac{2 e^{j\pi/4} e^{-j9t}}{2 e^{-j\pi/4} e^{j4t}} \right]$$
$$\hat{n}(t) = 16 + 6 \left[\frac{e^{j(3t - \pi/4)}}{e^{j(6t - \pi/2)}} + \frac{e^{-j(3t - \pi/4)}}{e^{-j(6t - \pi/2)}} \right] + 4 \left[\frac{e^{j(4t - \pi/4)}}{e^{j(4t - \pi/4)}} + \frac{e^{-j(4t - \pi/4)}}{e^{-j(4t - \pi/4)}} \right]$$

$$n(t) = 16 + 12 \cos(3t - \pi/4) + 8 \cos(6t - \pi/2) + 4 \cos(4t - \pi/4)$$

Date

$$\hat{n}(t) = 16 + \left[\frac{6 e^{-j\pi/4} e^{j3t}}{4 e^{j\pi/2} e^{-j6t}} + \frac{6 e^{j\pi/4} e^{-j3t}}{2 e^{-j\pi/4} e^{j9t} + 2 e^{j\pi/4} e^{-j9t}} \right] + \left[\frac{4 e^{-j\pi/2} e^{j6t}}{e^{j(6t-\pi/2)} + e^{-j(6t-\pi/2)}} \right]$$

$$\hat{n}(t) = 16 + 6 \left[\frac{e^{j(3t-\pi/4)}}{e^{j(3t-\pi/4)} + e^{-j(3t-\pi/4)}} \right] + 4 \left[\frac{e^{j(6t-\pi/2)}}{e^{j(6t-\pi/2)} + e^{-j(6t-\pi/2)}} \right]$$

$$+ 2 \left[\frac{e^{j(4t-\pi/4)}}{e^{j(4t-\pi/4)} + e^{-j(4t-\pi/4)}} \right]$$

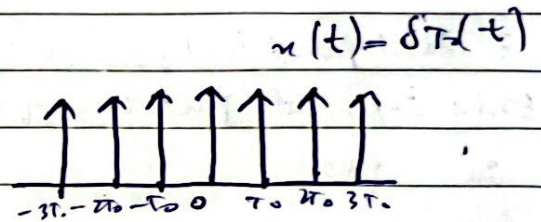
$$n(t) = 16 + 12 \cos(3t - \pi/4) + 8 \cos(6t - \pi/2) + 4 \cos(4t - \pi/4)$$

DAY-17

Impulse train and its fourier series:-

Impulse train is the key for analogue-to-digital conversion of signals.

the unit impulse train can be expressed as



$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

the exponential fourier series for this impulse train would be

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

where

$$D_n = \frac{1}{T_0} \int_{T_0} \delta_{T_0}(t) e^{-jn\omega_0 t} dt$$

over the interval $(-T_0, T_0) \rightarrow \delta_{T_0}(t) = \delta(t)$

$$\left(\int_{-\infty}^{\infty} \delta(t) dt = 1 \right) \quad \text{81-YAG}$$

Impulse is located at $t=0$. So using

$$\int_{-\infty}^{\infty} \delta(t) \delta(t) dt \quad [\text{sampling property}]$$

Date

$$= \phi(0) \int_{-\infty}^{\infty} \delta(t) dt$$

$$= \phi(0)$$

So

$$\int_{-T_0/2}^{T_0/2} \delta(t) e^{jn\omega t} dt$$

is evaluated at $t=0$ and becomes 1.

$$\text{So } D_n = \frac{1}{T_0}$$

FI-YACI

Now the exponential fourier series is

$$s_{T_0}(t) = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega t}$$

101

is the exponential fourier series

only amplitude plot is required as the phases

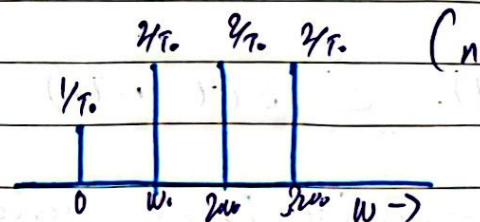
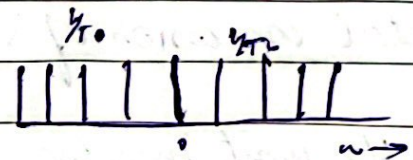
are zero.

Now,

$$C_0 = D_0 = 1/T_0$$

$$C_n = 2|D_n| = 2/T_0$$

$$Q_n = 0$$



and

$$s_{T_0}(t) = \frac{1}{T_0} [1 + 2(\cos \omega_0 t + \cos 2\omega_0 t + \cos 3\omega_0 t + \dots)] \quad \omega_0 = \frac{2\pi}{T_0}$$

is the trigonometric Fourier Series for impulse train $s_{T_0}(t)$.

DAY-16

Fourier Transform :-

By applying the limiting process, an aperiodic

Date

signal can be expressed as a continuous sum (integral) of everlasting exponentials.

$$x(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega t} d\omega$$

is known as the **Fourier integral**

$X(\omega) = F[x(t)]$ is the direct fourier transform of $x(t)$.

whereas

$x(t) = F^{-1}[X(\omega)]$ is the inverse fourier transform of $X(\omega)$

$$x(t) \Leftrightarrow X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

the transform $X(\omega)$ is the frequency domain specification of $x(t)$

the spectrum $X(\omega)$ can be plotted as a function of ω

$$X(\omega) = (X(\omega)) e^{j(\phi(\omega))}$$

$X(\omega)$ which is the fourier transform of $x(t)$ is more precisely the spectral density per unit bandwidth

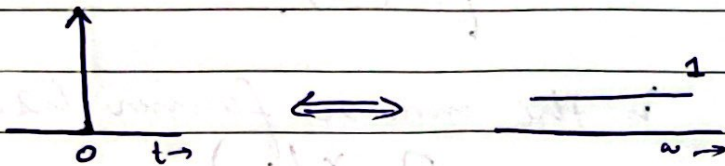
Question 1:-

Find the fourier transform of unit impulse function $\delta(t)$

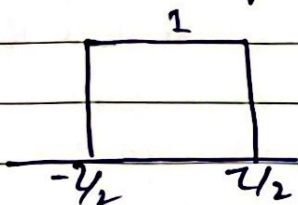
$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$= 1$$

**Question 2:-**

Find the fourier transform of $g(t) = \left\{ \text{rect} \left(\frac{t}{2} \right)^2 \right\}$



$$g(t) \longleftrightarrow G(\omega)$$

$$G(\omega) = ??$$

$$\text{we have } G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \text{rect} \left(\frac{t}{2} \right) e^{-j\omega t} dt$$

since $\text{rect} \left(\frac{t}{2} \right) = 1$ for $|t| \leq \frac{2}{2}$ and

since $\text{rect} \left(\frac{t}{2} \right) = 0$ for $|t| > \frac{2}{2}$

$$G(\omega) = \int_{-1/2}^{1/2} 1 \cdot e^{-j\omega t} dt = \frac{-1}{j\omega} \left(e^{-j\omega \frac{1}{2}} - e^{j\omega \frac{1}{2}} \right)$$

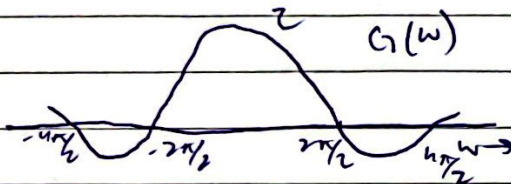
$$G(\omega) = 2 \frac{\sin \omega \frac{1}{2}}{\omega}$$

$$G(\omega) = 2 \frac{\sin \left(\omega \frac{1}{2} \right)}{\omega}$$

Date

$$G(\omega) = \tau \frac{\sin\left(\frac{\omega\tau}{2}\right)}{(\omega\tau/2)}$$
$$= \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$

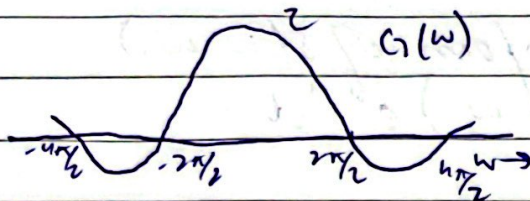
$$\operatorname{rect}\left(\frac{t}{\tau}\right) \iff \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$



Date

$$G(\omega) = \tau \frac{\sin\left(\frac{\omega\tau}{2}\right)}{(\omega\tau/2)}$$
$$= \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\operatorname{rect}\left(\frac{t}{\tau}\right) \iff \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$



DAY-19

Fourier Transform:

$$x(t) \iff X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\delta(t) \iff 1$$

Find the IFT of $\delta(\omega)$

$$F^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega$$

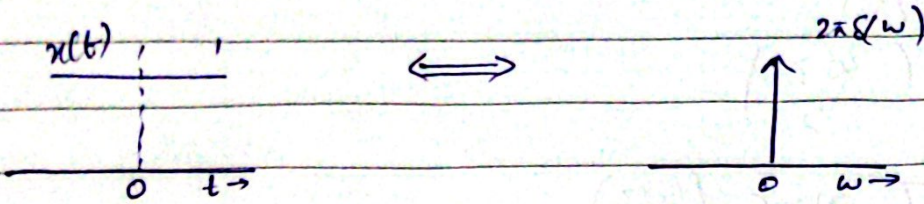
$$F^{-1}[\delta(\omega)] = \frac{1}{2\pi}$$

$$\frac{1}{2\pi} \iff \delta(\omega)$$

ω

Date

$$1 \Leftrightarrow 2\pi \delta(\omega)$$



Question:-

Find the inverse Fourier transform of $\delta(\omega - \omega_0)$

$$\begin{aligned} F^{-1}[\delta(\omega - \omega_0)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} e^{j\omega_0 t} \end{aligned}$$

PI-YAD

therefore

$$\frac{1}{2\pi} e^{j\omega_0 t} \Leftrightarrow \delta(\omega - \omega_0)$$
$$e^{j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega - \omega_0)$$

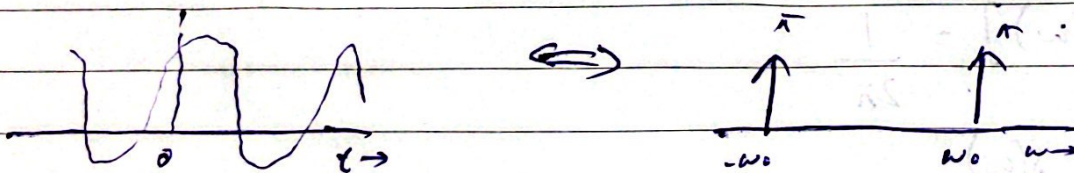
this result shows spectrum of an everlasting exponential $e^{j\omega_0 t}$ is the single impulse of $\omega = \omega_0$.

Also

$$e^{-j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega + \omega_0)$$

Question:-

Find the Fourier transform of the everlasting sinusoid $\cos \omega_0 t$



Date

$$\cos \omega_0 t = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$$

$$e^{j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega - \omega_0)$$

$$e^{-j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega + \omega_0)$$

An everlasting sinusoid $\cos \omega_0 t$ can be synthesized by two everlasting exponentials $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$. Therefore, the Fourier transform of $\cos \omega_0 t$ consists of only two components of frequencies ω_0 & $-\omega_0$.

Question:

Find the Fourier transform of $e^{-at}u(t)$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t) e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \frac{-1}{a+j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty}$$

But

$$|e^{-j\omega t}| = 1$$

If $a < 0$, as $t \rightarrow \infty$ $e^{-(a+j\omega)t} = e^{-at} e^{-j\omega t} = \infty$ (growing exponential)

If $a > 0$ $x(\omega) = \frac{1}{a+j\omega}$

Expressing $a+j\omega$ in polar form as

Date

$$\sqrt{a^2 + \omega^2} e^{-j \tan^{-1}(\frac{\omega}{a})}$$
$$x(\omega) = \frac{1}{\sqrt{a^2 + \omega^2}} e^{-j \tan^{-1}(\frac{\omega}{a})}$$

$$|x(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}} \quad \angle x(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$

